

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE BIOLOGY

Paper 2

Monday 2 June 2025

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



Answer **all** questions in the spaces provided.

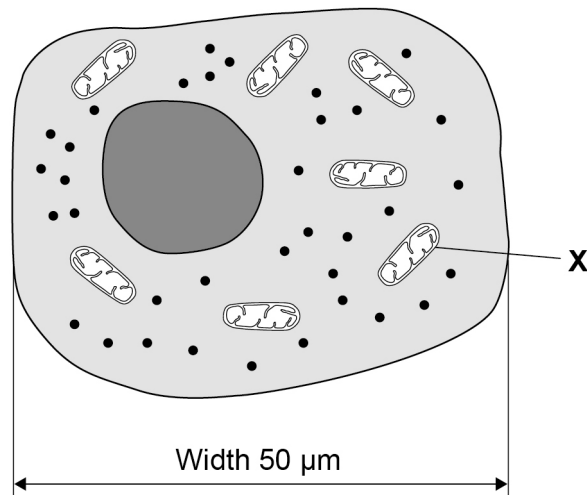
Do not write
outside the
box

0 1

All living things are made of cells.

Figure 1 shows a human liver cell.

Figure 1



0 1 . 1

What is part **X** in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Chloroplast

☐

Cytoplasm

☐

Mitochondrion

☐

0 1 . 2

Label the nucleus on **Figure 1**.

[1 mark]



0 1 . 3 What is the function of the nucleus in a cell?

[1 mark]

Tick (✓) **one** box.

To control the cell

☐

To make sugars

☐

To store food

☐

0 1 . 4 The liver cell in **Figure 1** has a real width of 50 µm.

1 mm = 1000 µm

Calculate the magnification of the image in **Figure 1**.

[4 marks]

Use the equation:

$$\text{magnification} = \frac{\text{width of image of liver cell}}{\text{real width of liver cell}}$$

Give your answer to the nearest whole number.

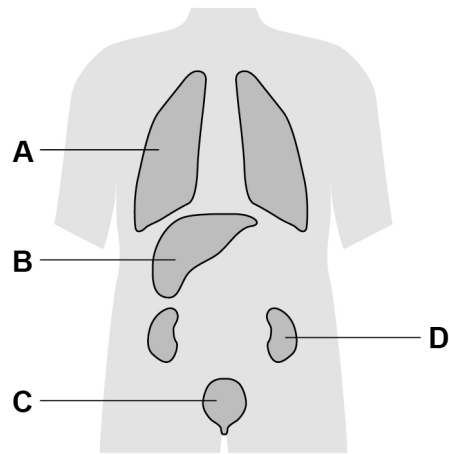
Magnification = _____

Turn over ►



Figure 2 shows parts of the human body.

Figure 2



The liver is an important organ in the body.

0 1 . 5 Which part is the liver in **Figure 2**?

[1 mark]

Tick (✓) **one** box.

A	☐	B	☐	C	☐	D	☐
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0 1 . 6 Which **two** are functions of the liver?

[2 marks]

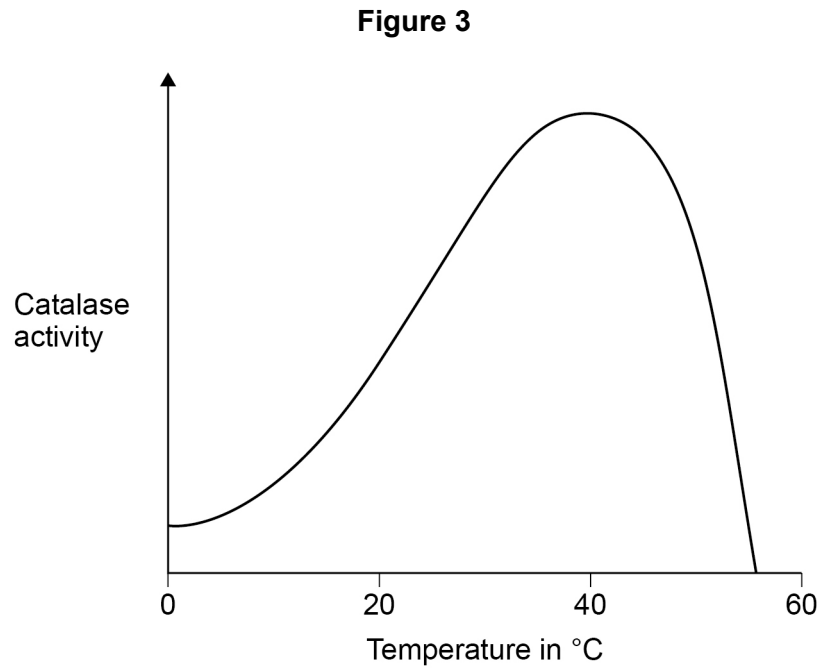
Tick (✓) **two** boxes.

To detoxify poisons	☐
To exchange gases	☐
To make bile	☐
To produce antibodies	☐
To release testosterone	☐



0 1 . 7 Liver cells contain the enzyme catalase.

Figure 3 shows catalase activity at different temperatures.



Describe the trend in catalase activity as temperature increases.

[2 marks]

0 1 . 8 Explain why catalase is **not** active above 55 °C.

[2 marks]

Turn over ►



0 1 . 9

Digestive enzymes are needed to break down large molecules into small molecules.

Table 1 shows information about digestive enzymes.

Complete **Table 1**.

[3 marks]

Table 1

Enzyme	Organ that produces the enzyme	Large molecule broken down	Small molecules produced
Amylase	salivary gland	starch	
	small intestine	lipids	fatty acids and glycerol
Protease	stomach		amino acids

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17



0 2

Substances are moved through plants.

0 2 . 1

Which structure is a plant organ?

[1 mark]

Tick (✓) **one** box.

Epidermis

☐

Leaf

☐

Xylem

☐

0 2 . 2

Plant roots absorb water from the soil.

What else do plants absorb through their roots?

[1 mark]

Tick (✓) **one** box.

Chlorophyll

☐

Mineral ions

☐

Sunlight

☐

0 2 . 3

Plants lose water from their leaves through tiny openings.

What is the name of the tiny openings on leaves?

[1 mark]

Tick (✓) **one** box.

Alveoli

☐

Stomata

☐

Valves

☐

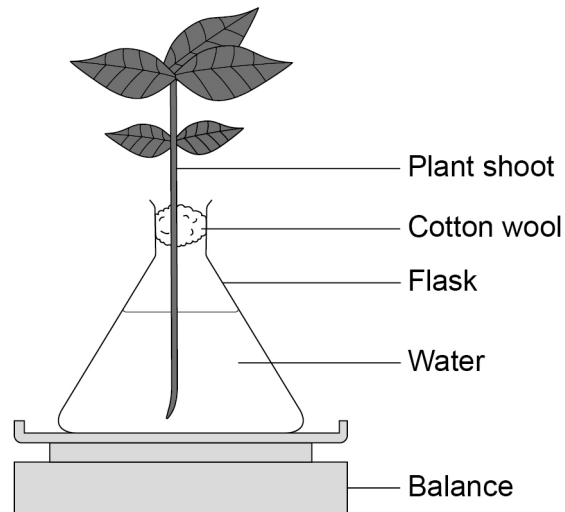
Turn over ►



A student investigated the effect of temperature on water loss from a plant shoot.

Figure 4 shows the apparatus used.

Figure 4



This is the method used.

1. Set up the apparatus shown in **Figure 4**.
2. Record the mass of the apparatus at the start.
3. Leave for 3 hours at 20 °C.
4. Record the loss of mass of the apparatus.
5. Repeat steps 2 to 4 at 40 °C and at 60 °C.

0 2 . 4 Give **two** variables the student kept the same in the investigation.

[2 marks]

1 _____

2 _____

0 2 . 5 What was the independent variable in the investigation?

[1 mark]



Table 2 shows the results.

Table 2

Temperature in °C	Mass of water lost in grams
20	1.2
40	2.3
60	4.8

0 2 . 6 Complete the sentence.

[1 mark]

Choose the answer from the box.

decreases

stays the same

increases

As temperature increases the mass of water lost _____.

0 2 . 7 Complete the sentence.

[1 mark]

Choose the answer from the box.

circulation

division

evaporation

The process that causes water loss is _____.

Question 2 continues on the next page

Turn over ►



0 2 . 8 Plants move substances into and out of cells by active transport and diffusion.

Describe **two** differences between active transport and diffusion.

[2 marks]

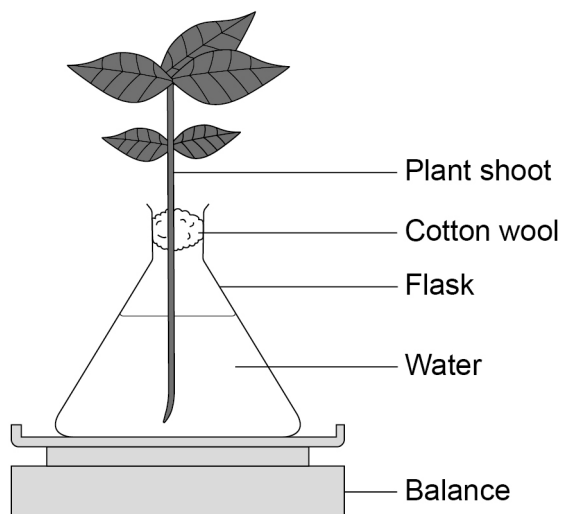
1 _____

2 _____



0 2 . 9 Figure 4 is repeated below.

Figure 4



Describe a method to investigate the effect of total surface area of leaves on water loss from a plant shoot.

Use the apparatus in **Figure 4**.

[4 marks]

14

Turn over for the next question

Turn over ►



0 3

This question is about the control of blood glucose concentration.

0 3 . 1

Which organ produces insulin?

[1 mark]

Tick (✓) **one** box.

Mouth

☐

Pancreas

☐

Stomach

☐

A person who did **not** have diabetes ate a meal high in sugar.

0 3 . 2

Explain the effect that eating a meal high in sugar would have on the person's blood glucose concentration.

[2 marks]

0 3 . 3

Explain how the body controls blood glucose concentration after eating a meal high in sugar.

[3 marks]



0	3	.	4
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Explain how high blood glucose concentration affects the water content of red blood cells.

[3 marks]

Question 3 continues on the next page

Turn over ►



Obesity is a risk factor for the development of Type 2 diabetes.

Drug **K** is a new drug used to treat people with obesity.

Scientists investigated the effect of drug **K** on patients with obesity.

Each patient was given one tablet of the same type each week for 64 weeks.

The tablet had:

- drug **K**
- or
- no drug.

A tablet with **no** drug is called a placebo.

0 3 . 5 The investigation used 1961 patients randomly divided into two groups.

One group received drug **K** and the other group received the placebo.

The ratio of those receiving drug **K** to those receiving the placebo was 2 : 1

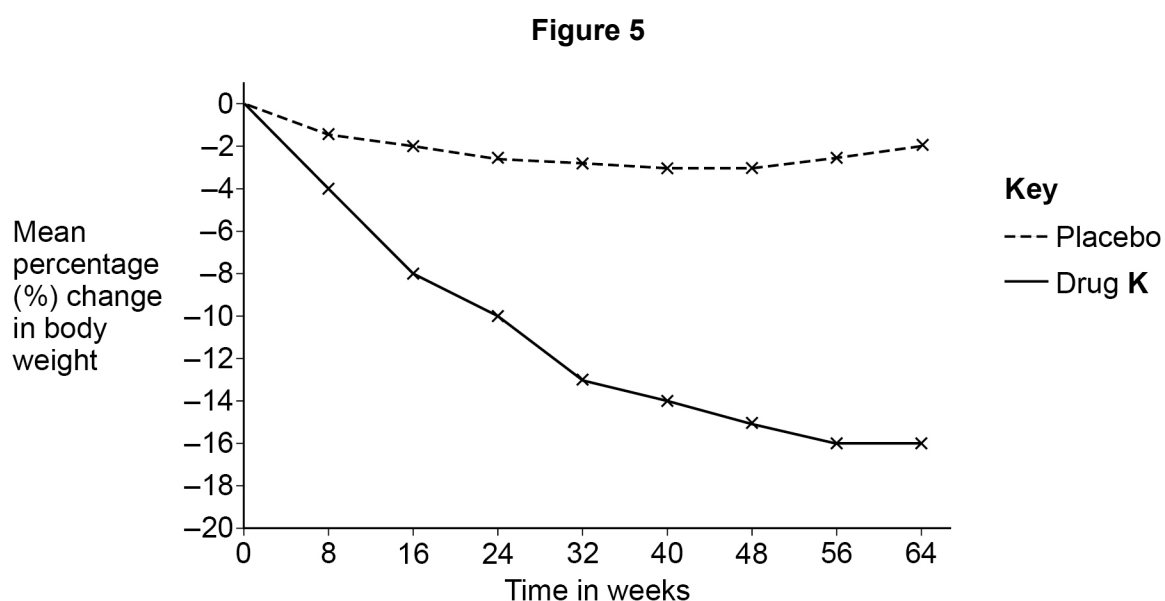
Calculate the number of patients who received drug **K**.

[3 marks]

Number of patients who received drug **K** = _____



Figure 5 shows the results of the investigation.



0 3 . 6

Estimate the mean percentage change in body weight for a patient receiving drug **K** at 12 weeks.

Use **Figure 5**.

[1 mark]

0 3 . 7

Give **two** conclusions that can be made from the results.

Use data from **Figure 5**.

[3 marks]

1

2

Question 3 continues on the next page

Turn over ►



0 3 . 8 The investigation continued until 128 weeks.

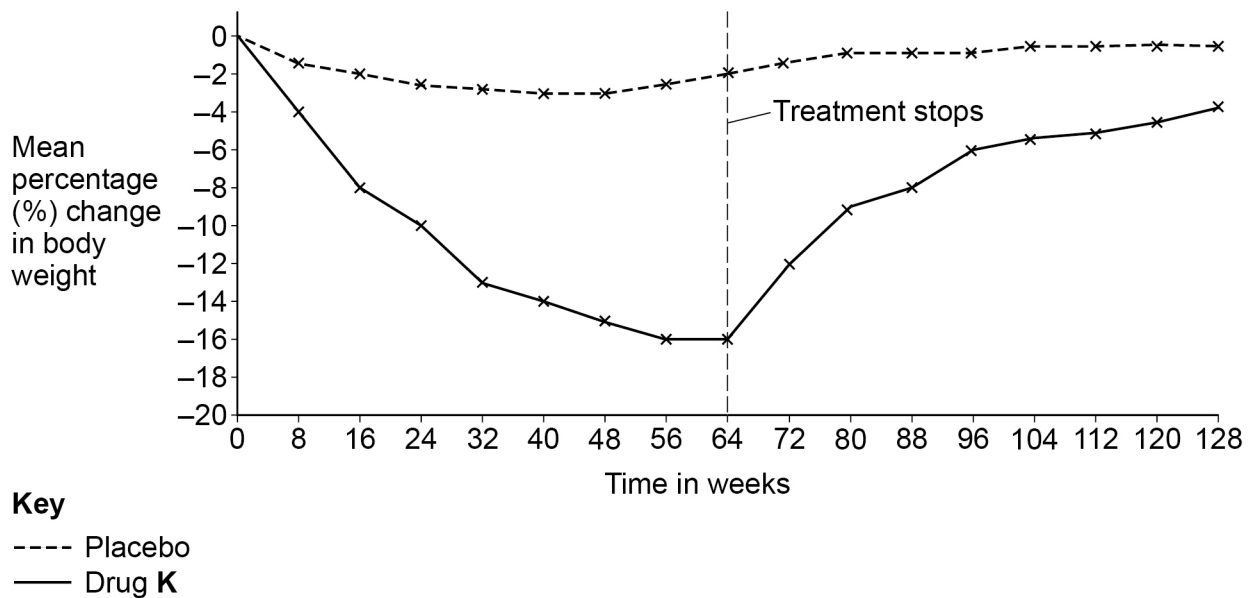
After 64 weeks, the patients did not receive any tablets.

Figure 6 shows both the results for the first 64 weeks as shown in **Figure 5** and the results from 64 to 128 weeks.

The patients:

- were over 18 years old
- were obese
- did **not** have diabetes
- controlled the energy intake in their diet
- did 150 minutes of exercise each week
- kept a record of their diet and exercise.

Figure 6



[6 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

22

Turn over ►

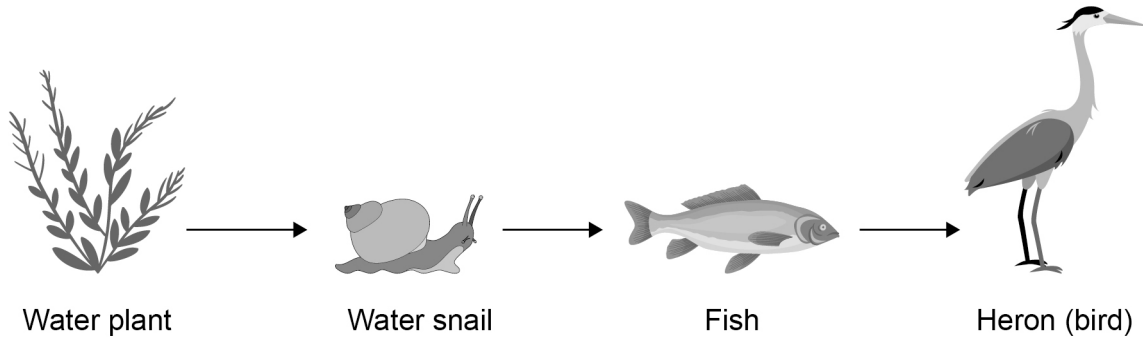


0 4

Many organisms live in water.

Figure 7 shows a food chain for a river.

Figure 7



0 4

1

Energy for all food chains comes from light.

Describe how the energy from light is transferred to the water snail.

[2 marks]

0 4

2

Only 10% of biomass is transferred from the water snails to the fish.

Give **one** way biomass is lost between water snails and fish.

[1 mark]



0 4 . 3 Sewage entering rivers causes eutrophication.

Eutrophication reduces the oxygen concentration in the water.

Explain the effect of eutrophication on the population of herons in the pond.

[2 marks]

Question 4 continues on the next page

Turn over ►



An increase in human population leads to an increase in sewage production.

Sewage can carry parasites that infect humans causing sickness and diarrhoea.

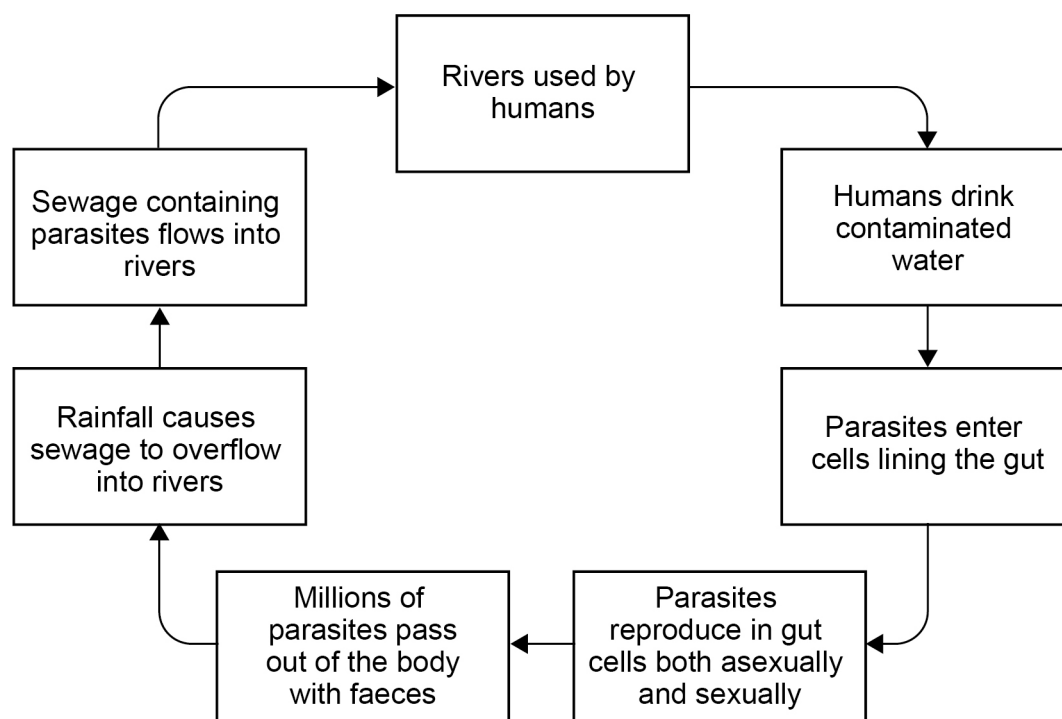
Diarrhoea is watery faeces.

Cryptosporidium is a single-celled parasite that infects humans.

Cryptosporidium parasites can survive in acids and in chlorine.

Figure 8 shows how *Cryptosporidium* can infect humans.

Figure 8



0 4 . 4

Suggest how **two** adaptations of *Cryptosporidium* parasites help them to survive and continue to infect humans.

[2 marks]

1 _____

2 _____



0 4 . 5

Suggest **one** reason why there are more cases of *Cryptosporidium* infection in hotter weather.

[1 mark]

Another human parasite is the malarial parasite.

0 4 . 6

Give **one** way the malarial parasite is **similar** to the *Cryptosporidium* parasite.

[1 mark]

9

Turn over for the next question

Turn over ►



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0 5

Sea turtles are animals that live in the ocean.

Figure 9 shows a sea turtle.

Figure 9



0 5

. 1

Sea turtles have adaptations that enable them to survive in the ocean.

Table 3 shows some adaptations.

Tick (✓) **one** box in **each** row to identify the type of adaptation shown by each feature. **[3 marks]**

Table 3

Feature	Type of adaptation		
	Behavioural	Functional	Structural
Body shape			
Migrate thousands of kilometres to find beaches where eggs can be laid			
Maintain body temperature above cold water temperature			

Question 5 continues on the next page

Turn over ►



Figure 10 shows information about reproduction in sea turtles.

Figure 10

- Adult turtles migrate thousands of kilometres to reach beaches to lay their eggs.
- Female turtles lay hundreds of eggs in sand on the beach.
- Many females lay eggs on the same beach.
- The females cover up and hide the eggs but give no other parental care.
- New baby turtles make sounds before leaving the egg. The sounds allow them to all leave the eggs together.
- Baby turtles innately crawl towards the water.
- Baby turtles take shelter among floating plants in the ocean.
- Only 6% of baby turtles which leave the eggs survive the first year.

0 5 . 2 Baby turtles use sound signals to communicate when to leave the eggs.

Give **two** other types of signals animals can use to communicate.

[2 marks]

Do **not** refer to sound in your answer.

1 _____

2 _____



0 5 . 3

Describe how the behaviour of sea turtles increases their chances of reproductive success.

[3 marks]

Use **Figure 10**.

0 5 . 4

Global warming is causing the decline of sea turtle populations.

Describe how global warming may reduce sea turtle populations.

[6 marks]

You should use information from **Figure 10** and your own knowledge.

Turn over ►

The place where one turtle lays its eggs is called a nest.

Scientists monitor turtle populations by counting their nests.

Table 4 shows data collected for one beach.

Table 4

Year	Number of nests
2004	3800
2006	3000
2008	2400
2010	1750
2012	1400
2014	1100

0 5 . 5 Describe the trend shown in **Table 4**.

[1 mark]

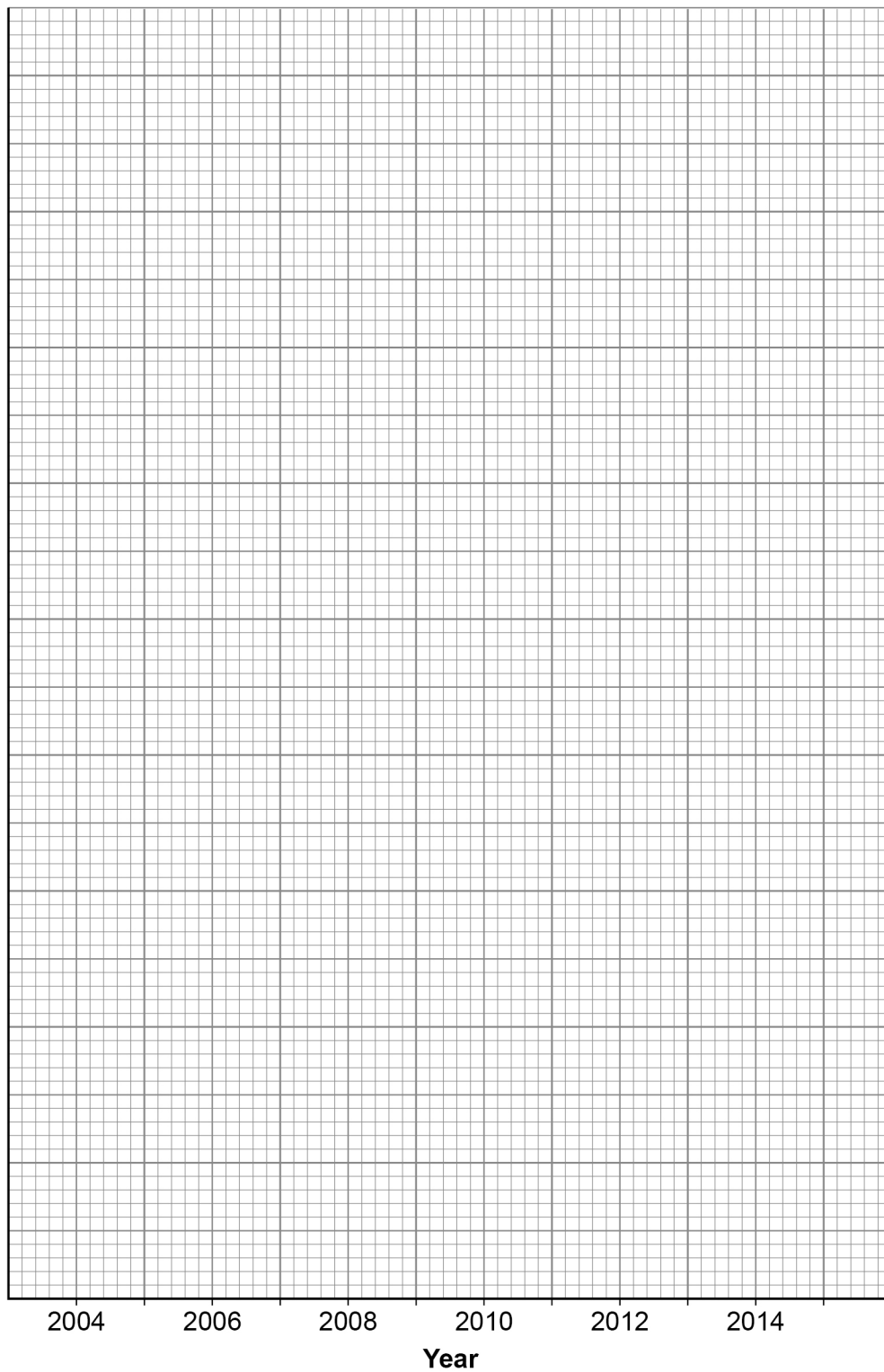
0 5 . 6 Complete **Figure 11**.

You should:

- use a suitable scale for the *y*-axis
- label the *y*-axis
- plot the data from **Table 4**
- draw a line of best fit.

[5 marks]



Figure 11**Turn over for the next question****Turn over ►**

0 6

Algae are able to photosynthesise.

Student **A** investigated the effect of light intensity on the rate of photosynthesis in algal balls.

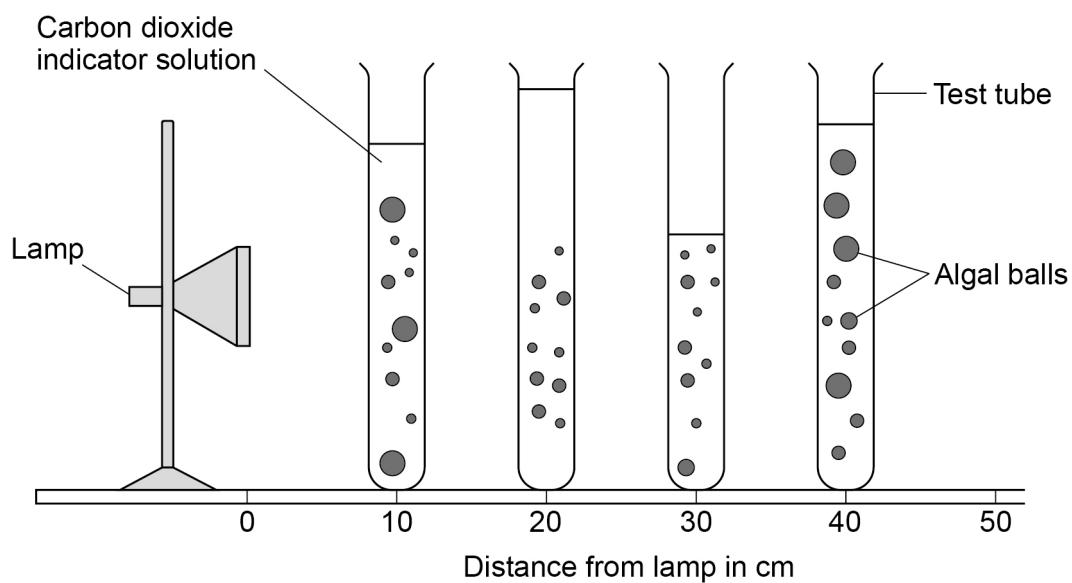
Algal balls are made from clear jelly containing many algal cells.

This is the method used.

1. Place test tubes at 10 cm, 20 cm, 30 cm and 40 cm from a lamp.
2. Add 10 algal balls to each test tube.
3. Add some carbon dioxide indicator solution to each test tube.
4. Add 1g of sodium hydrogen carbonate to each test tube to release carbon dioxide.
5. Turn on the lamp.
6. After 30 minutes, record the colour of the indicator in each tube.

Figure 12 shows the apparatus used.

Figure 12



0 6 . 1 The method did **not** give valid results.

Suggest **two** improvements the student could make to the method to gain more valid results.

Give a reason why each change you described would improve the method.

[2 marks]

Improvement _____

Reason _____

Improvement _____

Reason _____

0 6 . 2 The student identified the bulb in the lamp as a possible hazard.

Bulbs can break and cause injury to students.

Give **one other** risk of using a bulb.

Describe how to reduce the risk.

[2 marks]

Risk _____

How to reduce the risk _____

Turn over ►



Table 5 was given to students to show the colours of the indicator at different carbon dioxide concentrations.

An indicator changes colour in different concentrations of a chemical.

The carbon dioxide indicator was red at the start of the investigation.

Table 5

Concentration of carbon dioxide	Colour of indicator
very high	yellow
high	orange
normal	red
low	pink
very low	purple

0 6 . 3 Student **A** repeated the investigation using a valid method.

Table 6 shows the results.

Table 6

Distance from lamp in centimetres	Colour of indicator at start	Colour of indicator after 30 minutes
10	red	purple
20	red	purple
30	red	pink
40	red	red



Student **A** concluded:

‘The rate of photosynthesis increases as light intensity increases.’

Explain how the results in **Table 6** support the student’s conclusion.

[2 marks]

0 6 . 4

Student **B** wanted to investigate the effect of different colours of light on the rate of photosynthesis in algal balls.

Give **one** way student **B** could change the colour of the light.

[1 mark]

0 6 . 5

Student **B** did **not** use a carbon dioxide indicator to measure the rate of photosynthesis.

Suggest **one other** way Student **B** could measure the **rate** of photosynthesis.

[1 mark]

8

END OF QUESTIONS



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